Stack & Queue

→ Stack :- A stack is a linear data structure that follows the Last In, First Out (LIFO) principle, meaning that the last element added to the stack is the first one to be removed.

→ Basic operation on stack :

1.) Push - Adds an element to the top of the stack.
2.) Pop - Removes the top element from the stack.
3.) peek - Returns the top element without removing it.
4.) isEmpty - checks if the stack is empty.
5.) isFull - checks if the stack is full (for a fixed-size stack).

→ TOS (Top of stack) - TOS refers to the index or pointer that points to the topmost element in a stack.

(i) Initialized to -1 (indicating an empty stack).
(ii) Increases when elements are pushed.
(iii) decreases when elements are popped.
(iv) Used to access the top element (peek operation).

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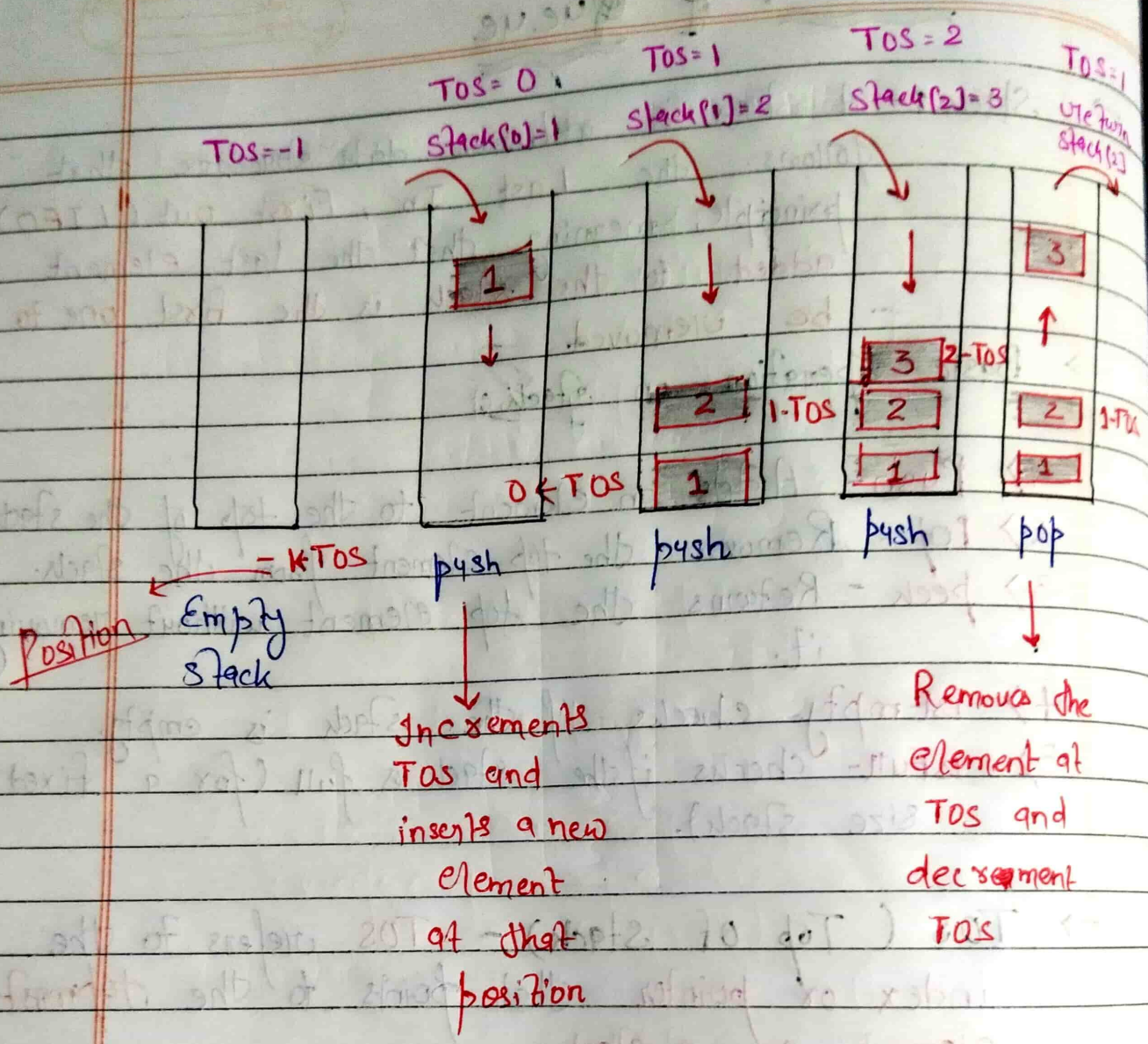

→ Implementation of stack in C:

A stack can be implemented using:

- array { Fixed size, static memory allocation }
- Linked List { Dynamic memory allocation }

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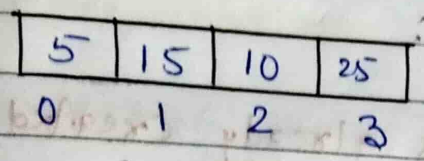
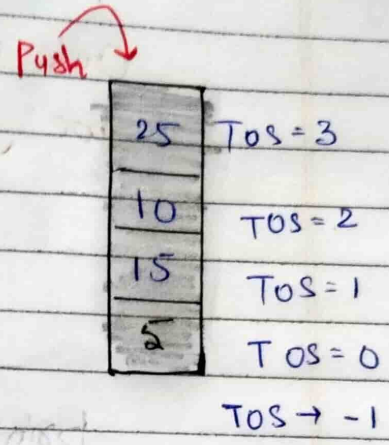
→ Stack implementation using array :-

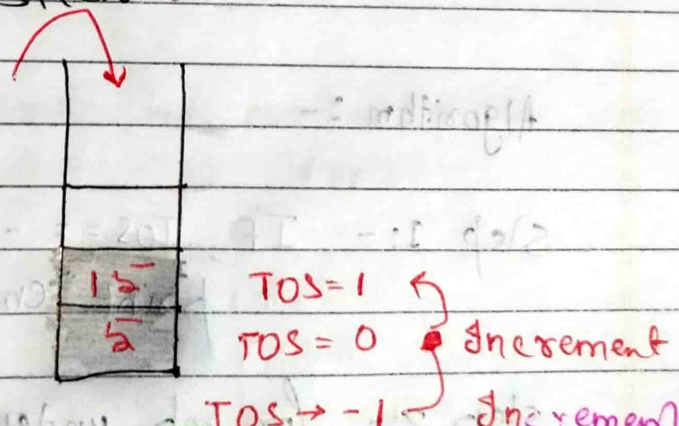
It means, to make an array and implement stack on it.



Array



(i) Creating a Stack :



Algorithm :-

Step 1:- set max 100

2: set tos = -1

3: if (n>max)

print "Stack overflow"

4: else,

for loop under condition i<n;

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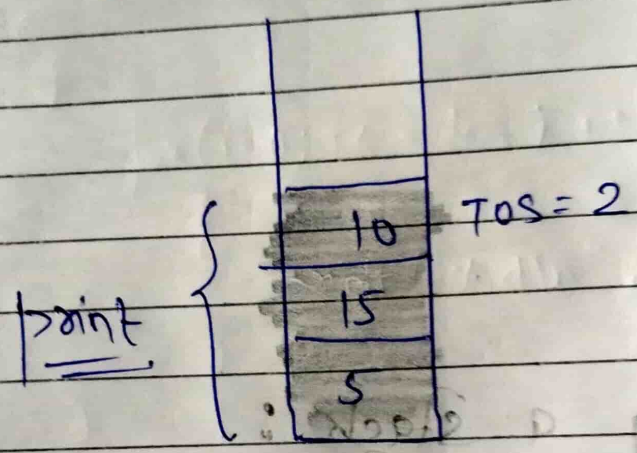
4.1.> set stack[++tos] = val

5.> Exit.

(ii) display stack:

Our stack is already created, we just have to display.



Algorithm :-

Step 1:- IF TOS == -1
print "empty list"

step 2:- for loop under condition i<=tos

2.1.> print stack[i]

step 3:- Exit.

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→ // C program to create & display ~~linked~~ stack using array.

```
#include<stdio.h>
#define max 100        // maximum size of array
int stack[max], tos=-1;

void creating_stack()
{
    int val, i, n;
    printf("enter no. of element to insert\n");
    scanf("%d",&n);

    if (n>max)
    {
        printf("stack overflow, cannot insert more
         than max element");
    }
  else
    {
        for (i=0; i<n; i++)
        {
            printf("enter data\n");
            scanf("%d",&val);
            stack[++tos] = val;
        }
    }
}
```

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```
void display()
{
    int i;
    if(tos==-1)
    {
        printf("empty stack");
    }
    else
    {
        for(i=0; i<=tos; i++)
        {
            printf("%d\n", stack[i]);
        }
    }
}

int main()
{
    creating_list();
    display();

    return 0;
}
```

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